

MailGuard AI Forensic Report

Case MG-26C84FA62A8B | generated 2026-10-01T08:35:19+00:00 (UTC)

Raw message SHA-256: 26c84fa62a8bdfd6527c0c63da01f97ce988f2f37a3d59097e5785fad18827be

BLOCK

Verdict class: Fraud

Probability of fraud: 0.9999

Total log-odds of fraud: +8.93 (itemised in section 3)

1. Message summary

From (display name)	HDFC Bank Accounts
From (address)	accounts@hdfc-verify.example
Reply-To	hdfc.recovery.desk01@gmail.com
To	finance@victim.example
Subject	URGENT: revised payment instructions, account change
Date	2026-09-18T03:41:12+00:00
Message-ID	<20260918034112.44718.9c2f@hdfc-verify.example>

2. Evidence table

One row per signal, ordered by contribution. Points are log-odds contributions to the verdict above: they sum, with the interaction terms and the base rate in section 3, to the total log-odds. A signal marked "abstain, not scored" could not run; that is missing data, not a clean result, and it still carries points, because the model scores an absence as an absence rather than as a zero.

	Signal	Score	Status	Points	Evidence
x2	Identity	0.91	ok	+3.28	cousin domain of hdfc.in, brand label 'hdfc' embedded in an unrelated domain, display name claims 'hdfc', reply-to free mail at gmail.com, first contact
x5	Intent	0.86	ok	+2.75	urgency (subject): urgent; bank detail change: all future payments should be made to; payment redirection: our old account is frozen; secrecy: keep this confidential
x7	Attachment and URL	0.78	ok	+2.65	Invoice_HDFC_44718.xlsm: macro document (vbaProject.bin present inside the OOXML zip); link text says www.hdfc.in but points to xn--hdfc-4n1e.in; shortened link, not resolved
x3	Infrastructure	0.72	ok	+1.15	hdfc-verify.example, registered 4 days before receipt, no MX records, hosting ASN (Bulk Relay Hosting BV)
x4	Sender Baseline	abstain, not scored	abstain	+0.60	no baseline: 0 prior messages, 5 required
x6	Origin and Geo	0.15	ok	+0.21	tier 2 provider_bounded: 203.0.113.19 (AS200000, Bulk Relay Hosting BV, NL)
x1	Authentication	0.05	ok	+0.04	SPF pass, DKIM pass (d=hdfc-verify.example, s=mailer01), DMARC pass

3. Model arithmetic

interaction x2 with x5	+1.72
interaction x2 with x7	+0.64
interaction x6 with x7	+0.07
interaction x1 with x6	+0.01
base rate (intercept)	-4.20
total, log-odds of fraud	+8.93
probability after calibration	0.9999

Fusion: documented heuristic priors standing in for the untrained EBM, standard model; first contact, no sender baseline.

4. Origin attribution

TIER 2: attribution stops at the infrastructure. The boundary host and its operator are established; the person or machine behind them is not. Do not read the country above as the sender's location.

Tier	2 - provider_bounded
Boundary IP	203.0.113.19

ASN	AS200000
ISP / operator	Bulk Relay Hosting BV
Country	NL (location of the boundary host, NOT of the sender)
Notes	Boundary is a shared outbound relay. The relay's own inbound hop claims 10.8.0.14, an RFC1918 address written by a host we do not control, so the true origin is not established. Attribution stops at the relay operator.

5. Relay trace

Hops 0 to 2 were recorded by infrastructure under our control and can be relied on. Hop 3 and everything below the rule were written by hosts we do not control and may be fabricated.

hop 0 trusted from mx-in-02.victim.example [198.51.100.12] by mail.victim.example 2026-09-18T03:41:20+00:00
from mx-in-02.victim.example (mx-in-02.victim.example [198.51.100.12]) by mail.victim.example with ESMTPS id 4c1f9a; Fri, 18 Sep 2026 03:41:20 +0000

hop 1 trusted from filter-07.victim.example [198.51.100.7] by mx-in-02.victim.example 2026-09-18T03:41:18+00:00
from filter-07.victim.example (filter-07.victim.example [198.51.100.7]) by mx-in-02.victim.example with ESMTP id 9b23ee; Fri, 18 Sep 2026 03:41:18 +0000

hop 2 trusted from smtp-out-19.bulkrelay.example [203.0.113.19] by filter-07.victim.example 2026-09-18T03:41:15+00:00
from smtp-out-19.bulkrelay.example (smtp-out-19.bulkrelay.example [203.0.113.19]) by filter-07.victim.example with ESMTPS id 71ad02; Fri, 18 Sep 2026 03:41:15 +0000

TRUST BOUNDARY - everything below this line is attacker writable

hop 3 attacker writable from localhost [10.8.0.14] by smtp-out-19.bulkrelay.example 2026-09-18T03:41:12+00:00
from localhost (unknown [10.8.0.14]) by smtp-out-19.bulkrelay.example with ESMTPA id 0f44b1; Fri, 18 Sep 2026 03:41:12 +0000

6. Campaign linkage

Campaign id	CMP-C1E7EEDB1C
Messages in campaign	2
First seen	2026-09-18T03:41:12+00:00
Last seen	2026-09-18T03:41:12+00:00
Shared asn	as200000
Shared boundary ip	203.0.113.19
Shared country	nl
Shared dkim selector	mailer01
Shared reply to	hdfc.recovery.desk01@gmail.com
Shared url domain	bit.ly, xn--hdfc-4n1e.in

7. Chain of custody

Raw message SHA-256	26c84fa62a8bdfd6527c0c63da01f97ce988f2f37a3d59097e5785fad18827be
Sealing entry hash	3421a6d7e0c1cecdb662dda010cedb20da1c0a9a2842869cb8700935f7554428
Current chain head	2776428b763f7d6ffec05a9cfbf2524d1d96c3e01297dedc91b726cc5995fe16
Sealed at	2026-10-01T08:35:19+00:00
Ledger entries	4 for this case, 4 in the whole ledger
Chain verification	intact, every link verified
Ledger	C:\Users\laryan\AppData\Local\Temp\mailguard-report-demo-zv19k6ww\demo_ledger.jsonl

Each ledger entry stores the SHA-256 of the entry before it. Altering any earlier record changes its hash and therefore breaks every hash after it, so the chain cannot be edited silently; the first broken link is reported by index. The sealing entry records the hash of the message as received, taken before any parsing, which is what makes it evidence about the message rather than about this tool.